Manual no 3

Title

DCT-Based Robust and Imperceptible Digital Image Watermarking

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Aim

To implement a Discrete Cosine Transform (DCT) based method to embed a robust and imperceptible digital watermark into an image for protecting proprietary data.

Objective

1. To understand the concept of digital watermarking.
2. To apply DCT for frequency-domain watermark embedding.
3. To embed watermark data into mid-frequency DCT coefficients.
4. To achieve robustness against compression and noise.
5. To maintain imperceptibility of the watermarked image.

Robustness Against Compression and Noise

Robustness against compression and noise means the ability of a digital watermark to **remain detectable and intact even after the image undergoes compression (like JPEG- Joint Photographic Experts Group) or is corrupted by noise**.

Prerequisites

- Basic knowledge of:
    - Digital Image Processing
    - Grayscale images and pixel intensity
    - Frequency domain transforms

- Understanding of:
    - Discrete Cosine Transform (DCT)
    - Image block processing (8×8 blocks)
- Tools:
    - Python
    - OpenCV (cv2)
    - NumPy

Theory

Digital Watermarking

Digital Watermark in Image Processing

A digital watermark in image processing is hidden information embedded inside an image that is used to identify ownership, authenticate content, or protect proprietary data without noticeably affecting the image's visual quality.

Purpose of Digital Watermarking

- **Copyright protection**
- **Ownership verification**
- **Content authentication**
- **Prevention of unauthorized copying**
- **Tamper detection**

Key Characteristics

1. **Imperceptibility – The watermark should not degrade image quality.**
2. **Robustness – It should resist common image processing attacks (compression, noise, filtering).**
3. **Security – Unauthorized users should not detect or remove it.**

4. **Capacity – Amount of data that can be embedded.**

Types of Digital Watermarks

- **Visible watermark: Logo or text clearly visible (e.g., TV channel logo).**
- **Invisible watermark: Hidden and detectable only by algorithm.**

Based on Domain

- **Spatial domain: Direct pixel modification (e.g., LSB).**
- **Frequency domain: Transform-based (DCT, DWT, FFT).**

Example

When a photographer uploads images online, a hidden watermark containing their name or ID is embedded. Even if the image is copied or compressed, ownership can still be proven by extracting the watermark.

Digital watermarking embeds hidden information (watermark) into digital media to protect ownership and authenticity. A good watermark must be:

- **Imperceptible** – should not degrade visual quality
- **Robust** – resistant to compression, noise, filtering, cropping

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Why DCT for Watermarking?

- DCT converts an image from spatial domain to frequency domain.
- JPEG compression is DCT-based → watermark survives compression.
- Human eyes are less sensitive to changes in **mid-frequency components**.

DCT Frequency Bands

- **Low frequency**: Important image content (don’t modify)

- **High frequency**: Easily removed by compression/noise
- ✅ **Mid frequency**: Best place to embed watermark

DCT Formula

2D DCT

For an 8×8 block:

$$D(u,v) = \frac{1}{4}C(u)C(v)\sum_{x=0}^{7}\sum_{y=0}^{7} f(x,y)\cos\left(\frac{(2x+1)u\pi}{16}\right)\cos\left(\frac{(2y+1)v\pi}{16}\right)$$

Where:

$$C(k) = \begin{cases} \frac{1}{\sqrt{2}}, & k = 0 \\ 1, & k > 0 \end{cases}$$

Watermark Embedding Formula

Let:

- D_1, D_2= selected mid-frequency coefficients
- $W \in \{0,1\}$= watermark bit
- α= embedding strength

Embedding Rule

$$\begin{cases} D_1 = D_1 + \alpha, & \text{if } W = 1 \\ D_2 = D_2 + \alpha, & \text{if } W = 0 \end{cases}$$

Algorithm (Embedding Process)

1. Read input image and convert to grayscale.
2. Divide image into 8×8 blocks.
3. Apply DCT to each block.
4. Select two mid-frequency coefficients.
5. Embed watermark bit by modifying coefficient values.
6. Apply inverse DCT (IDCT).
7. Reconstruct the watermarked image.

Numerical Example

Given 8×8 DCT Block (partial values)

$$D = \begin{bmatrix} 120 & 15 & 10 & 8 \\ 14 & 20 & \mathbf{6} & \mathbf{5} \\ 9 & 7 & 4 & 3 \end{bmatrix}$$

Selected coefficients:

- $D_1 = 6$
- $D_2 = 5$

Let:

- Watermark bit $W = 1$
- Embedding strength $\alpha = 2$

Embedding

$$D_1 = 6 + 2 = 8$$

Modified block:

$$D = \begin{bmatrix} 120 & 15 & 10 & 8 \\ 14 & 20 & \mathbf{8} & \mathbf{5} \\ 9 & 7 & 4 & 3 \end{bmatrix}$$

After IDCT → spatial block

➞ Visual change is **imperceptible**

Pseudocode: DCT-Based Digital Image Watermarking

START

Read input image in grayscale mode

Convert image pixel values to floating point

Set block size = 8

Set embedding strength α

Set watermark bit (0 or 1)

Get image height (H) and width (W)

Create a copy of the original image for watermarking

FOR i = 0 to H-1 STEP block_size

  FOR j = 0 to W-1 STEP block_size

    Extract 8×8 block from the image

    Apply DCT to the 8×8 block

    IF watermark_bit == 1 THEN

      Modify mid-frequency DCT coefficient at position (4,3)

      Increase coefficient value by α

ELSE

Modify mid-frequency DCT coefficient at position (3,4)

Increase coefficient value by α

END IF

Apply Inverse DCT to the modified block

Replace original block with watermarked block

END FOR

END FOR

Save the final watermarked image

STOP

Conclusion- Write in your own words.